

Sound online growth of a typed *isa* lattice from noisy LLM extraction, through candidate elimination made noise-tolerant by a localized-blame admission gate

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Abstract

A language model knows a great deal and asserts it without restraint; a knowledge base asserts only what it can defend, but everything must be written into it by hand. Systems that couple the two currently choose between two failure modes. Either the symbolic side never learns — every sort, every synonym remains an authoring cost. Or the model writes into the base itself, and the base gradually absorbs its *world-plausibility*: plausible-sounding facts that nothing supports. The textbook case of this second mode is NELL, the longest-running self-growing knowledge-base experiment: years of autonomous growth, plausible-but-false facts admitted with no correction channel, and a drift that neither co-training nor human checks ever stopped.

This article presents the third option, and measures it. The host structure is a **typed *isa* lattice**: a hierarchy of sorts ("a marble is a ball, a ball is a round thing") on which tasks declare their requirements. Three kinds of units must be able to enter it along the way: a **slot restriction** (which sorts a task role accepts), an ***isa* edge** (a parent-child link between sorts), and a **surface alias** (a synonym of a declared vocabulary word). The need is precise: grow these three units online, from the noisy extractions of a small local language model, without absorbing the model's ontology. The instrument is a single admission rule: **a piece of evidence is admitted for a unit only if its success or failure is uniquely attributable to that unit — by structural provenance or by counterfactual ablation — and verifies against the declared oracle**. One gate, three grains. The theoretical justification takes three steps. One: candidate elimination, the classical algorithm for learning such restrictions, is provably intolerant to noise — a single false negative expels the right answer forever. Two: the noise of an LLM pipeline is **incompetence noise** — one-sided (it only manufactures false failures) and competence-correlated (rare cases fail systematically) — precisely the kind that statistical noise models do not cover. Three: localizing blame replaces the noisy query over a whole conjunction with a clean query over the single responsible literal; the residual noise on admitted negatives then reduces to the confounded-episode case — bounded by a defeasible two-tier envelope, recoverable by retraction, never zero.

The evidence follows three levels. In a deterministic laboratory (no model, exact pre-registered expectations), the gate halves over-generalization without ever refusing a good task, while the control that admits every failure self-seals on rare cases. Under live conditions, with an embedded 27-billion-parameter model as the sole organ of world knowledge: 300/300 tasks against 245/300 for

the model alone, the model’s deficit concentrated exactly where one must refuse, retract a default, or follow the ontology in depth; and zero false edges, zero false aliases admitted on permuted streams, where the ungated variant absorbs the model’s ontology and then answers wrongly with no correction channel left — NELL’s drift, reproduced in miniature then blocked, at both grains. On the third-party benchmark DeFAB, the typed path scores 34/35 (30/35 of them with no model call at all) against 30/35 for the direct model, and every loss of the direct model is an over-general cut — the error class the gate forbids by construction. A reproduction across nine local models (four families, three quantizations, two architectures and two size brackets) shows that the decider, the gate and the fail-closed refusal all generalize; only coverage tracks extraction capability. There remains the economics: what retrieval pipelines re-pay in context on every call, this system compiles once into a typed, versioned library, auditable to the episode — knowledge accumulates outside the context window, and it is the gate that makes this accumulation safe. None of the bricks is new; the composite is: an LLM that extracts, a lattice that decides, and a gate that lets the lattice grow without drifting.

Keywords: selectional restrictions; version spaces; candidate elimination; *isa* lattice; defeasible reasoning; knowledge-base drift; blame attribution; neurosymbolic systems; online learning; LLM extraction.

1. Introduction

1.1 Who decides, and who learns

Large language models are excellent organs of world knowledge and rather poor arbiters of that knowledge. The example that will serve throughout this article is a placement riddle: a prose instruction describes an object ("the yellow ball") and a set of typed holes (round, square, star); the task is to say which hole the object goes into — or to refuse, if none accepts it. Ask a good model to place a ball: it answers correctly almost every time. Ask it to place a *pyramid* when none of the offered holes accepts it: the same model produces a fluent, plausible, wrong answer — a canonical pyramid does indeed have a square base, and world-plausibility is precisely what the model optimizes. Turning on *reasoning* (the thinking budget of recent models, which we will write *rb*) does not repair this case; it makes it more eloquent.

The failure is not ignorance. It is that the model follows the world it has read, not the specification it was given — with and without reasoning (we measure this in §6.3).

The classical remedy is to let a symbolic structure decide: extract typed facts from the prose, match them deterministically against a declared ontology, and refuse in a typed way when nothing matches. This division of labor is old and sound, and recent neurosymbolic pipelines implement it well (§2). It does, however, carry a structural cost that its own practitioners name explicitly; the knowledge base is a *manual bottleneck*. This "manual bottleneck" means that every sort, every subsumption edge, every surface synonym must be written by hand. The obvious, naive escape — letting the model write the missing edges itself — is one of the great documented catastrophes of the field: NELL. NELL was the longest-running self-growing knowledge-base experiment, and it drifted despite co-training and periodic human correction, because nothing in its admission path could tell a fact the world supports from a fact the model finds plausible.

This article is about the admission path. We keep the division of labor (the model extracts, the lattice decides) and add the third capability this division seemed to forbid: **the lattice grows**

online, from the model’s noisy proposals, without absorbing the model’s ontology. The instrument is thus a single admission rule, applied at three grains of the structure.

1.2 The claim

*A defeasible generalization drawn from an LLM episode — a slot restriction, an isa edge, or a surface alias — is admitted **if and only if** its success or failure is uniquely attributable to that single unit (by structural provenance or by counterfactual ablation) **and** verifies against the declared oracle.*

We call this rule the **localized-attribution admission gate**. The theoretical content of the article is the following chain, developed in §4 and priced in §5:

1. Our hypothesis class consists of conjunctions of cuts over a finite, fixed *isa* lattice — a *slot* is a typed role of a task schema (the object to place, the receiving hole); a *cut* is the set of sorts a slot admits, downward-closed in the lattice. Since this class has finite elasticity, it is therefore identifiable in the limit from positive examples alone. Thus, negative evidence is *not* a logical necessity; it buys convergence speed and control of the generalization frontier. Note, moreover, that this guarantee covers *cut selection* within a given lattice; the *growth* of the lattice itself belongs to a different regime, which we will describe in §4.2.
2. The negative evidence an LLM pipeline actually produces is noisy in the worst way: one-sided (a valid sort fails because the model was incompetent on it, not because it is wrong) and competence-correlated (rare, hard sorts fail systematically). This is not random classification noise; statistical tolerance results do not transfer, and classical candidate elimination is provably fragile to a single error of this kind.
3. Localization repairs this by construction rather than statistically: an episode failure is therefore admitted as a negative for slot *i* only when the execution contract localizes the violated atom on *i*. Thus one can convert a noisy membership query over the whole conjunction into a clean query over the responsible literal; the noise on *admitted* negatives then stops being the model’s incompetence rate and becomes the false-admit rate of the admission predicate itself — what §4.5 names the *episode confound*, bounds with a two-tier envelope, and makes recoverable by retraction. Let us note it right away: this rate is not zero, and no "noise-free case" rate is restored in the PAC sense (§4.4 states exactly what is bought).
4. The same rule, applied symmetrically, governs:
 - positive credit (a success credits only the slots whose atoms were actually exercised);
 - *isa*-edge admission (an edge proposed by the model is *optimistically mounted* — that is, tried immediately rather than held pending proof —, verified, and credited only on episodes whose verdict cannot be due to anything else);
 - and surface-alias admission (an out-of-vocabulary word joins a synonym *ring* — that is, the alias ring attached to a sort’s key — only if it is load-bearing under counterfactual ablation: the verdict passes with it, fails without it).

One gate, three grains — a unification of invariant, not of implementation.

Everything else in the system is deliberately old:

- Fillmore’s role-typed slots;
- Mitchell’s version spaces;
- an *isa* lattice as subsumption order;
- defeasible edges labeled with source and confidence;

- and retraction by truth maintenance.

The contribution is the composite and its gate — and the fact that the whole loop runs on a single small local model.

1.3 How the evidence is organized

The article follows its evidence through three levels, from most controlled to most exposed:

- **Theory (§4).** What is learnable, under what noise, and why localization is the principled remedy — with the known results this rests on, and the exact points where they fail to transfer.
- **The deterministic laboratory (§5).** The learning rule isolated from any model: four arms on identical permuted streams, 126/126 exact pre-registered checks. This is where the gate’s value is *priced* — what it buys, what it costs, and what the unsound alternatives do in its place.
- **Existence under live conditions (§6–§7).** The full circuit with an embedded 27-billion-parameter model as extractor and proposer: constancy over fresh instances, domains and volume (§6), the drift baselines at both grains (§6.5–§6.6), and external validity on a third-party benchmark with a machine-verifiable oracle (§7).

We state the scope honestly right now: the live results are existence results, on declared toy lattices, at existence-level N in several cells, with the oracle circularity the setup implies — which we flag wherever it exists. What the three levels establish together is not that the mechanism is complete, nor that its constants are final — but that it exists, that it is priced, and that it survives contact with a third-party oracle.

2. Related work, and the mechanism delta

The bricks used here are decades old; we therefore claim none of them:

- the defeasible inheritance and retraction date back to Reiter;
- version spaces and candidate elimination were formalized by Mitchell;
- selectional restrictions as learnable cuts over a hierarchy, by Resnik’s information-theoretic model over WordNet;
- and noise-tolerant inductive logic programming is likewise mature (Popper; the MDL induction from noisy data of Hocquette et al.).

The rest of this section, however, situates the composite described in this article among its neighborhoods and must therefore — because they are well populated — name for each the exact *delta* rather than an empty cell. Five neighborhoods are to be visited — online LLM+solver pipelines, the defeasance evaluations, self-growing knowledge bases, learning selectional restrictions, and version spaces under noise — and a table recapitulates the position at the end of the section.

LLM + symbolic solver, online. The first neighborhood is also the strongest current line of work: coupling an LLM front-end to a logic-programming engine. Four landmarks stand out:

- Logic-LM and Logic-LM++ translate the problem into solver input and self-refine on solver errors — a repair of *formulation*, per problem, with no persistent memory and no blame;
- the closest mechanism twin is the self-correcting LLM-as-ASP-programmer pipeline (arXiv:2604.27960). It natively expresses defaults and exceptions (ASP is non-monotonic) and is explicit about its own bounds:
 - one-shot correction, per session;

- not a learning system;
- no blame attribution;
- no defeasible revision of *learned* knowledge;
- the s(CASP)-socialbot line and ProSLM have native defeasance and typed abstention — over a hand-written base and rules, a bottleneck their authors name;
- the sound-and-complete neurosymbolic reasoner of arXiv:2507.09751 delivers the fidelity face of our claim.

It offers abstention on inconsistent or uncertain cases and fidelity to a declared theory, with the LLM housed *inside* the semantics’ interpretation function — but the theory remains static: nothing in it learns a rule, an edge or an alias.

The defeasance evaluations. The second neighborhood builds no system: it measures. Three evaluations there document, at volume, the same failure:

- DeFab (arXiv:2606.18557) — the one we reuse as an external oracle in §7;
- DEFREASING (NAACL 2025);
- the generics-and-defaults study (arXiv:2508.13718).

The documented failure: current LLMs fail defeasible *overrides* — they do not retract a default when a modifier defeats it. These evaluations are foils in the strict sense: they establish the failure our demonstration exercises.

Because this cell is both occupied and well measured, *defeasance is this article’s demonstration, never its contribution.*

Self-growing knowledge bases. The third neighborhood is that of the problem itself — bases that grow on their own. NELL (introduced in §1.1) is its textbook case, positive and cautionary at once; three lines live there today:

- NELL’s direct heir exists: DySECT (arXiv:2603.06915, from the NELL team).
It continuously grows a self-expanding base from LLM-extracted triples, in a closed extraction $\leftrightarrow$ knowledge loop — with no admission gate anywhere in the write path; this is exactly the configuration whose drift §6.5–§6.6 measure, and the contrast this article instruments;
- recent taxonomy-induction work (SC-Taxo; the LLMs4OL challenge series) treats structural inconsistencies and semantic misalignment with coherence filters — global consistency, not per-unit localization;
- provenance-favoring graph completion (TGComplete, arXiv:2606.15833) is the closest parent of an admission gate: it verifies a candidate through a lightweight retrieval loop and *abstains* when textual support is missing.

The delta with this closest parent is precise: its verification is documentary support at completion time; there is no optimistic mount, no per-unit localized credit/blame, and no defeasible retraction of what has been admitted.

Learning selectional restrictions. The fourth neighborhood is the very object we learn, and its history takes two steps:

- learning theory stopped at the corpus-batch setting of the 1990s (Resnik; class-based estimation);
- the neural era first turned the question into *probing*, then reopened it on the learning side: arXiv:2011.02417 shows that a fine-tuned model forms robust selectional generalizations after one or two instances of a novel word.

That is a real neighbor of our question, with a clean delta: the learning there lives in the weights — neither auditable, nor attributable, nor retractable.

Online learning of restrictions *from failures*, with localized blame, externalized into an inspectable structure, remains — as far as our review reaches — without an occupant.

Version spaces under noise. The last neighborhood is the closest to the title, and it would be dishonest to keep it quiet: candidate elimination has had noise-tolerant variants for twenty-five years —

- Sebag’s disjunctive version spaces (one space per positive, classification by vote);
- Hirsh’s generalized version spaces (bounded inconsistency absorbed by merging);
- rough version spaces (approximation bounds);
- robust k-DNF via belief merging.

This whole family makes candidate elimination noise-tolerant by **relaxing the boundary** — statistically, by vote, approximation or merging.

Our mechanism is orthogonal: the boundary stays strict, and it is the **admission channel** that filters — a negative enters only when it carries a structural attribution to a single literal, verified. Likewise, the per-unit counterfactual ablation (β , §4.4) joins the recent wave of counterfactual/Shapley credit assignment in multi-stage pipelines — used here not as post-hoc explanation, but as an *admission criterion* for structural units.

The position, in one table:

Twin	Has	Lacks
Self-correcting LLM-as-ASP-programmer (2604.27960)	extraction $\rightarrow$ online solver, correction loop, natively non-monotonic	persistent learning · blame · defeasible revision of learned rules
Sound-and-complete neurosymbolic (2507.09751)	soundness, abstention, fidelity (LLM in the interpretation)	any learning (static theory) · credit/blame
ProSLM / s(CASP)-socialbot	extraction $\div$ application, native defeasance	hand-written base (named bottleneck) · learning
NELL / DySECT (2603.06915)	online growth at scale, closed loop	a sound admission path (drift)
TGComplete (2606.15833)	verify-and-abstain at completion, provenance	optimistic mount · per-unit credit/blame · retraction
Noise-tolerant VS (Sebag; Hirsh; rough VS)	candidate elimination under noise	the boundary is relaxed; nothing is attributed or audited
Popper / MDL-noise ILP (Hocquette et al.)	noise-tolerant induction, with exceptions	any link to an online LLM intake

Novelty is therefore claimed at the **mechanism delta**, not the cell: (c) online growth of lattice edges from LLM extraction **with** a localized-attribution admission path — where the self-growing line (NELL, DySECT) writes without a gate and the verifying line (TGComplete) verifies without learning; and (c′) localization **as the admission criterion** making candidate elimination tolerant to incompetence noise — where the noisy-version-space family relaxes the boundary instead of filtering the channel.

What our review found nowhere is the *noise model* itself — one-sided, competence-correlated — named as the reason statistical tolerance does not transfer (§4.3).

We claim neither typed refusal, nor extract-then-decide pipelines, nor defeasible reasoning: all three exist. We claim the gate, and the measured composite.

3. The system by example

This section walks the whole circuit once — from input prose to the two kinds of learning — so that the formal and experimental sections land on prepared ground. The running example is the placement riddle of §1.1 (an object described in prose, typed holes, place or refuse), the very family the experimental part then measures at volume. Eight panels follow in circuit order; each points to the section that carries its evidence, and every figure is generated from the recorded traces of the real runs (Appendix A).

The host system, in one paragraph. The substrate is a rule-driven knowledge-graph engine in which every unit of structure is a *concept*. This article needs only three pieces of it:

- the **concept-sorts** — the nodes of the *isa* lattice (sorts, categories, facets), whose parent-child edges *are* the subsumption relation, and on whose keys the synonym rings live;
- the **intake** — the typed extraction and its canonicalization barrier (panels 1–2);
- and the **admission gate**.

Epistemic status is a qualifier, never a type: every sort, edge or alias is either an *axiom* (authored) or *learned* — defeasible, labeled {**source**, **confidence**}, retractable. The same engine and the same discipline carry our companion system article [Braun 2026], which documents the rest of the substrate (learned methods with role-typed slots, method-pack export); figures F3, F7 and F8 illustrate this host context and **carry no claim of the present article**.

concept-method "compare" — post-canon shape [aggregate, aggregate, check, emit], discovered at discovery run:

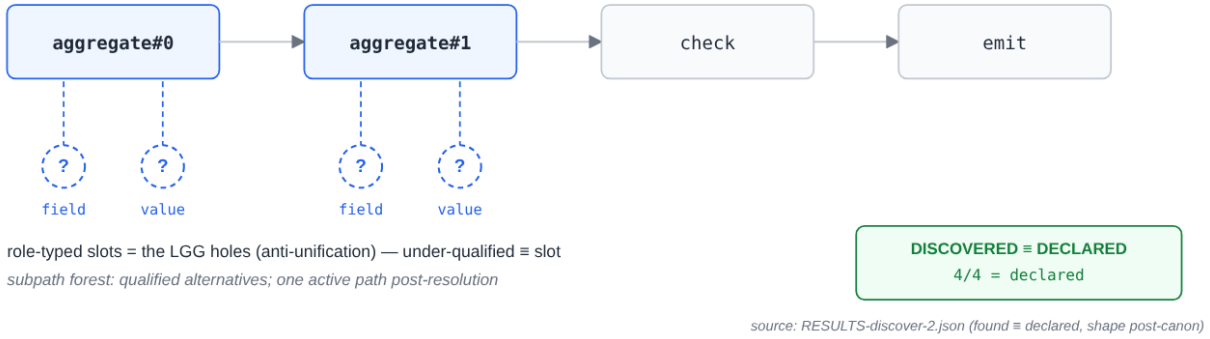


Figure F3 (host context, [Braun 2026]) — the anatomy of a concept-method: the post-canon shape of a discovered "compare" frame, with its role-typed slots — the holes of the LGG (the least general generalization, §4.1). Generated from RESULTS-discover-2.json.

Panel 1 — from prose to typed facts. An episode begins as paraphrased prose — the paraphrase is produced by the model itself, so the surface is never ours:

"Would you slide the sun-yellow ball into the circular opening?"

A dedicated extraction prompt returns typed facts, and only typed facts: an object of sort **ball** carrying the facet **color=yellow** (a *facet* is a transverse property — color, size, shape — that qualifies an object without being its sort); a set of holes {**star**, **square**, **round**}; a requested placement. Two disciplines apply at this boundary. First, the *typed-fact discipline*: everything downstream keys on discrete, typed facts — enums, identifiers, booleans — never on prose, so every decision is memoizable and replayable. Second, surfaces are extracted **verbatim**: the model reports the words the prose used ("circular opening") and is not allowed to canonicalize them — canonicalization is the system's job, and keeping it out of the model's hands is what will later make alias learning attributable (§6.6).

Panel 2 — the canonicalization barrier. Extracted surfaces are snapped onto the lattice's declared vocabulary: an exact match snaps; a *unique* containment snaps; everything else — including an *ambiguous* containment, such as a bare hypernym that several sorts contain — stays out-of-vocabulary (OOV), kept raw, fail-closed. OOV is not an error state; it is the honest channel by which unknown surfaces reach the learning machinery instead of silently corrupting the match (§6.4 shows what the ambiguous-containment rule prevents; §6.6 shows what OOV feeds).

model-paraphrased prose (real episode, durable memo):

You possess a deflated red football. Into which of the following openings should you place it: the star-shaped hole, the square hole, or the round hole?

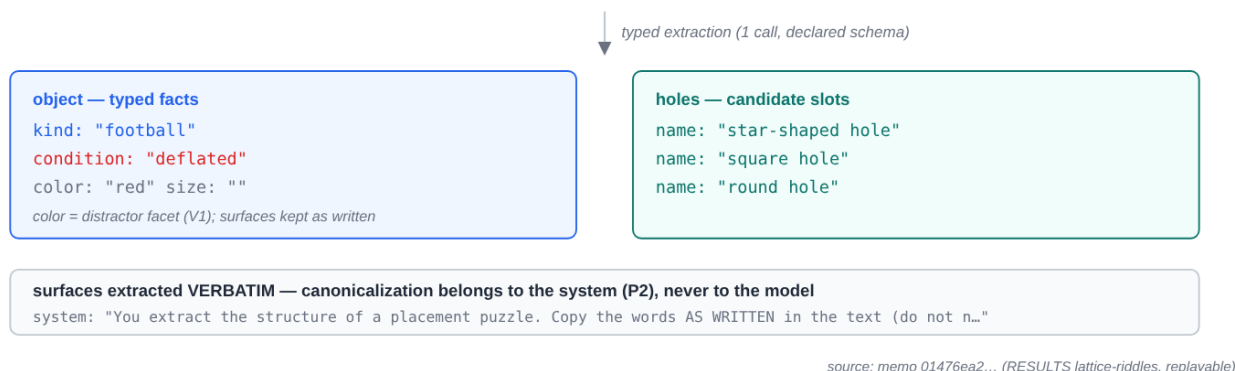
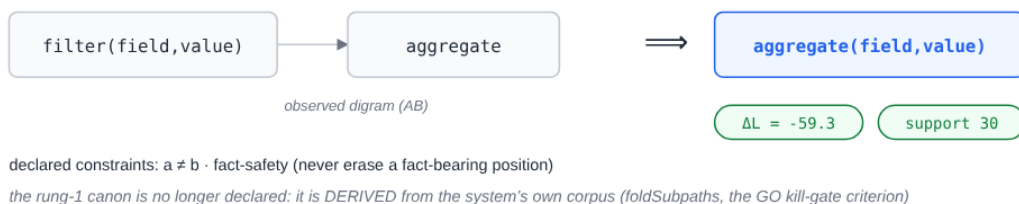


Figure F1 — a real episode (durable memo, replayable): the model-paraphrased prose, annotated, and its verbatim typed extraction — the sort, the defeating condition, the distractor facet, the holes.

the learned structural fold — the only digram admitted by MDL over the 23 captured shapes:



source: RESULTS-discover-2.json folds (foldSubpaths, MDL admission)

Figure F7 (host context, [Braun 2026]) — the same barrier, structural face: the digram fold [filter → aggregate] ⇒ aggregate(field,value), the only fold admitted by a minimum-description-length (MDL) criterion over the host system's captured shapes (ΔL = -59.3, support 30; seven other digrams rejected). The structural canon there is no longer declared: it is learned. No claim of the present article rests on this figure.

Panel 3 — the lattice, and a restriction as a cut. The declared ontology is a finite *isa* lattice of concept-sorts: `marble` $\sqsubseteq$ `ball` $\sqsubseteq$ `round-thing`, `die` $\sqsubseteq$ `cubic-thing`, with facets such as **shape** and **size** alongside. A *selectional restriction* for a slot is a **cut** through this lattice: the round hole accepts everything **round-thing** subsumes; a deeper riddle ("slide the sugar cube...") is unsolvable without the depth-2 chain `sugar` $\sqsubseteq$ `cube` $\sqsubseteq$ `cubic-thing` — which is exactly how the probes make the lattice load-bearing by construction (§6.2). Edges carry their epistemic qualifier: authored edges are axioms; learned edges are defeasible, `{source, confidence}`.

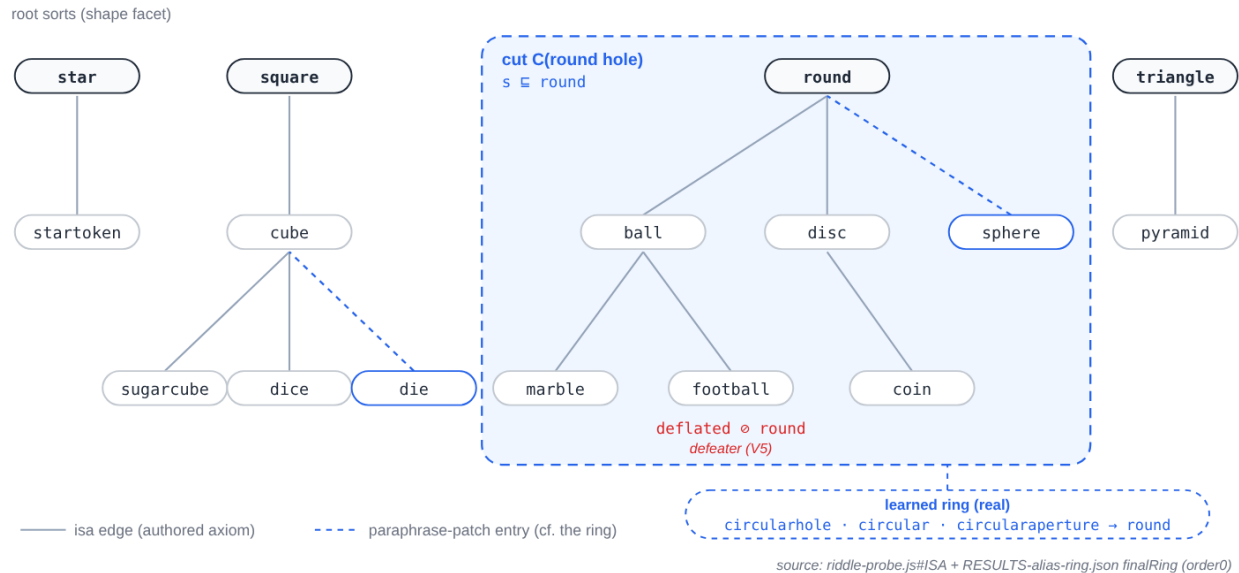


Figure F2 — the declared isa lattice of the shapes domain (read from the probe source), the round hole's restriction drawn as a cut, the defeater `deflated` $\otimes$ `round` (V5), and the synonym ring actually learned in the ring run (`circularhole · circular · circularaperture` $\rightarrow$ `round`), attached to the sort's key.

Panel 4 — the deterministic match, layered resolution. Matching a typed object against the holes' cuts is a deterministic lattice walk — zero model calls, a few microseconds, memoizable under the typed-fact discipline. When a cut admits the object, the task is **mounted**: the solution is instantiated on trial and then exercised — a *mount*, and it is its execution contract that will produce the credit or blame evidence. The resolution doctrine is layered, and this order is load-bearing: **the *isa* path is authoritative as soon as the sort is known; explicitly extracted facets serve only as a fallback for out-of-vocabulary sorts.** The reason is measured in §6: if explicit facets can override the lattice, the model's world knowledge leaks through the extraction ("a ball is round, I'll write `shape=round`") and the lattice silently stops being the decider (*figure F4, left panel*).

Panel 5 — the typed refusal. Give the system a pyramid and the holes `{star, round, crescent}`: no cut admits it. The answer is not a guess; it is a typed refusal — **impracticable**, with a structured hint naming the missing requirement ("no hole accepts sort `pyramid`; closest cuts: ..."). Returning a placement here is, by definition, a structural hallucination. §6.3 hardens this cell until the direct model has no world-plausible escape left — and it still fabricates a placement in 2 out of 6 instances, in both reasoning regimes, where the typed path refuses 6 out of 6.

Panel 6 — defeasance, and its vacuity guard. "Put the *deflated* football into the round hole." The condition `deflated` is extracted as a typed fact; a defeater edge `deflated` $\otimes$ `round` fires; the default conclusion is retracted, and — in the remap variant — re-derived to the flat slot

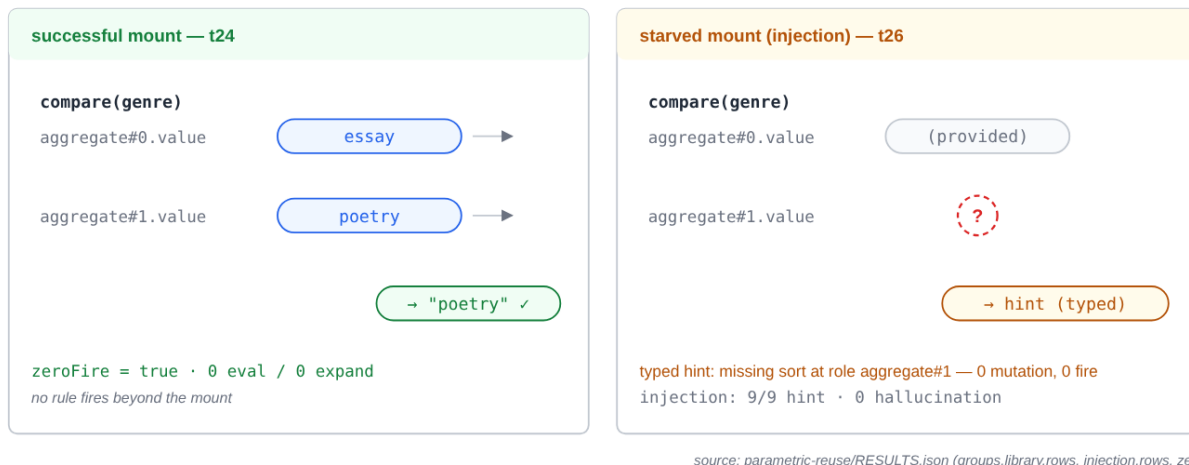


Figure F4 — two real episodes from the parametric probe: on the left the successful mount (t24 — the params placed into the role-typed slots, correct answer, zero firing beyond the mount); on the right the starved mount (t26, injection — the missing role produces a typed hint, never a plausible-looking answer: 9/9, 0 hallucinations).

where the deflated ball now fits. The direct model, at both reasoning budgets, pattern-matches *through* the modifier ("a ball is round") in the majority of instances (§6.3). The mechanism has the guard the demonstration needs: *benign* modifiers (wet, shiny, brand-new — surviving paraphrase as damp, gleaming, fresh) must defeat **nothing**, and defeat nothing (3/3 per domain) — the system discriminates defeaters; it has not simply learned "modifier $\Rightarrow$ refusal".

Panel 7 — learning a missing edge. To *ablate* the lattice is to deliberately remove the edges under a sort, simulating an incomplete ontology — the normal case of a deployed system. Do so, then give the system "put the trout into one of the enclosures": the strict lattice path now fails closed, and it is in this failure that learning begins. The model is asked for the missing placement; the proposal ($\text{trout} \sqsubseteq \text{fish}$, $\text{fish} \sqsubseteq \text{aquatic}$) is **optimistically mounted**, exercised, verified against the episode's oracle, and *credited only if the episode localizes the success on that edge* — no co-present unknown, no confounded verdict. Admission is provisional at first support; confirmation requires a second, de-confounded exposure. A later localized blame retracts it. This circuit recovers **39/39** ablated edges across the §6.2 probes (the no-match and defeasible cells ablate nothing by construction — there is no edge under a refusal), and — the point of the article — admits **zero** false ones on the same streams where the ungated variant absorbs the equivalent of six cells' worth (§6.5).

Panel 8 — learning a surface alias. Paraphrases manufacture vocabulary constantly: "die" arrives as "dice", "round" as "circular", "melted" as "liquefied". Every unknown surface lands OOV (panel 2) and becomes a proposal for the **synonym ring** of a lattice key. The gate here is counterfactual and per-unit: the alias is admitted only if the episode's verdict *passes with it and fails without it* — an ablation executed deterministically on the lattice-pure path, at zero model calls — then confirmed at support ≥ 2 over verified reuses. On the real stream, this admits exactly the six spec-true aliases the stream produces (including two nobody had planted — the paraphrase invented them, the gate caught them anyway), and refuses the model's spontaneous world-plausible snapping ($\text{aperture/cavity/hole} \rightarrow \text{round}$), which the ungated variant absorbs for good — two wrong downstream answers and a poisoned ring (figure F6, §6.6 — the ratchet timeline: *UNGATED drifting*, *GATED converting the same episodes into typed refusals and retractions*).

The sentence the example was built to earn. The slot restriction of panel 4, the *isa* edge of panel 7 and the alias of panel 8 are admitted by the *same* rule — unique attribution plus verification — realized by two mechanisms: structural provenance where the structure can name its unit, counterfactual ablation where it cannot. One gate, three grains (*figure F5, §4.4*). The rest of the article prices this sentence (§4–§5), then defends it under live conditions (§6–§7).

4. The formal layer

This section follows the questions in the order they arose while building the system:

- which objects, first (§4.1)?
- what is learnable over these objects, and with exactly what scope (§4.2)?
- that scope acquired, what noise breaks the classical learner one would want to use there (§4.3)?
- that noise named, why is localization the principled remedy, and what exactly does it buy (§4.4)?
- the remedy in place, what can "sound" still mean once its residual hole is admitted (§4.5)?
- and finally, does the same reasoning hold on the success side (§4.6)?

Each answer motivates the next question — each floor is the foundation of the next — and each subsection reopens on its question.

4.1 The objects

The first question: over which objects does everything play out?

To answer it, fix a finite lattice of sorts ($L, \sqsubseteq$): formally, a finite poset where multiple parents are allowed (joins are not necessarily unique, which will matter below). A vocabulary aside, before going further: *sort* is meant in the technical sense of many-sorted logics — the declared type of an object, heir to LOGIN's sort lattice (Ait-Kaci & Nasr 1986) — and, in the engine, these sorts are the concept-sorts under the `childConcepts` subsumption. Aside closed; the objects, then. A task schema exposes r role-typed slots. A **restriction** for slot i is a cut $C_i \subseteq L$; a sort s satisfies it iff $s \sqsubseteq c$ for some $c \in C_i$. A complete restriction is the conjunction over slots — a *monomial of sort cuts*. The language is deliberately capped at monomials (cuts per slot, possibly k-CNF above): learning arbitrary DNF over sorts is NP-hard (Pitt–Valiant), and the cap is checked at authoring time.

Learning maintains, per slot, $S_i = \text{LGG}$ of the verified positives — the least general generalization, computed by joins in L ; since joins are not unique in a multi-parent poset, S_i is a set of **parallel cuts**, collapsed only by evidence (parallel is safe; collapse is a collision). The G boundary of the version space is **never materialized**: for conjunctions it can be exponential even in trivial cases (Haussler), and for our class it is useless — S plus the exclusions *is* the hypothesis.

4.2 Learnability without negatives — and its exact scope

The objects in place, the question becomes: what is learnable over them, and with exactly what scope? **Scope first, because it separates the easy from the hard.** This whole subsection concerns *cut selection* within a **fixed** lattice; the *growth* of the lattice (edges, aliases — the contribution of the article) changes the hypothesis family mid-course, and the in-the-limit guarantee would have to be re-established for the augmented family — we do not do so. The growth layer

carries **no in-the-limit identification guarantee** in this article: its soundness is the recoverability of §4.5, and its evidence is empirical (§6.5–§6.6). This is consistent with §8, and it is said here so the theorem does not appear to cover what it does not cover.

Back to the fixed lattice, then. The class of per-slot cuts is **finite**, which gives it finite elasticity trivially; the general theorems (Angluin 1980; Wright 1989; Motoki–Shinohara–Wright 1991 — finite elasticity is preserved under finite unions and conjunctions) will only earn their place the day the *growing* family is treated. Consequently, the class is identifiable in the limit **from positive data alone**; Gold’s negative result does not apply — it bites superfinite classes, not this one. As an idealized reference: for the subclass with *one* literal per slot, $m = O((1/\varepsilon)(r \cdot d \cdot \ln b + \ln(1/\delta)))$ examples suffice in the i.i.d. regime (Haussler 1988) — linear in depth d , logarithmic in branching b over r slots. This formula calls for two pieces of honesty. On the one hand it uses $\ln|H| \approx r \cdot d \cdot \ln b$, valid for a single-ancestor cut per slot — the **parallel cuts** of §4.1’s multi-parent poset grow $|H|$ well beyond that. On the other hand its i.i.d. is not our online stream, permuted and competence-correlated. It serves as a landmark, never as a guarantee. With *informative* negatives, exact identification takes only $O(r \cdot (d+b))$ well-placed examples.

The design consequence deserves the emphasis, because it inverts the usual framing: **negative evidence is not a logical necessity for this class**. Positives identify in the limit on their own. What failure-derived negatives buy is *speed* and *frontier control* — how fast S stops over-generalizing, and whether it converges to the declared target or to something looser.

That is exactly what §5 measures: the LGG-only arm *converges* — to the wrong fixed point, too broad.

4.3 The noise that breaks candidate elimination

The learnable is delimited; the next question: what noise breaks the classical learner one would want to use here? To see it, start from the source. Where do negatives come from in an LLM pipeline? From a mounted restriction whose execution contract fails. That is a **membership query** (MQ) in Angluin’s sense, but degenerate: one-sided (only the “no” is informative, and only on the sorts that *arrive*), distribution-restricted (no arbitrary queries), with no equivalence oracle. The absence of EQ is benign — simulable by sampling. The noise is not.

The failure channel is **incompetence noise**: a *valid* sort gets labeled negative because the model botched the episode for reasons of its own, not because the sort violates the restriction. This noise is (i) one-sided — it only manufactures false negatives, which push the frontier *downward*, toward over-tightening — and (ii) **competence-correlated** — rare, hard sorts fail systematically, not at a fixed flip rate. Both properties matter:

- Classical candidate elimination is provably fragile to *any* mislabeled example: a single bad negative can permanently expel the target concept from the version space (Mitchell). Naive CE is unsound here — that is a theorem-grade statement, not a caution.
- Let us name the noise exactly, because the right library shelf depends on it. It is **one-sided classification noise with an instance-dependent rate**: zero on verified positives, rising with the sort’s rarity and difficulty, up to the agnostic regime on sorts the model fails systematically — a Massart–Nédélec-style profile, never a constant rate. One-sidedness is what saves the edifice: positives remain trustworthy, so the LGG-of-positives remains sound.
- This noise fits neither classical box. It is not *random*: the remedies for random classification noise (disagreement minimization at cost $\times 1/(1-2\eta)^2$; conjunction robustness via statistical queries) assume an example-independent rate, which competence correlation violates — **the**

statistical guarantees do not transfer.

- Nor is it *malicious* in the Kearns–Li sense: nothing adversarially corrupts the data channel, and invoking their impossibility bound literally would doom the gate itself, which reads the same channel. The real rescue, developed in §4.4, is therefore not a better noise model: it is the replacement of the noisy oracle by a deterministic one.
- K-corroboration (requiring K failures before admitting a negative) tests $P(\text{failure} \mid \text{sort})$ — it filters random noise and *lets through* systematic noise: K correlated failures of the same rare sort clear the bar and over-tighten anyway, at K times the cost, on sorts too rare to recur K times.

And one-sidedness has a dynamic consequence every design must face: **over-generalization self-corrects** (an S too broad will eventually cover a bad arrival, fail, and be re-cut), but **over-tightening is self-sealing** — a sort rejected in error is never mounted again, so the positive that would exonerate it is never collected.

The asymmetry is not cosmetic; it decides which errors are recoverable.

4.4 The gate

The noise is named; the question becomes: why is localization the principled remedy, and what exactly does it buy? The principled remedy is not a noise model but credit assignment:

***Admission rule.** A piece of evidence e from an episode is admissible for unit u iff $\text{attribution}(e) = \{u\}$ — the success or failure is uniquely attributable to u — and $\text{verify}(e)$ holds against the declared oracle.*

On the blame side, the mechanism is as follows: an episode failure becomes a negative for slot i only when the execution contract **localizes the violated atom** on i — all failing atoms point to the same role. A failure whose causes form a disjunction (several possible slots, or any co-present unknown, such as an unresolved OOV word) is **discarded**, never down-weighted.

The effect is structural. A noisy membership query over the whole conjunction becomes a clean query over the single responsible literal: on admitted negatives, the noisy oracle (the model) is *replaced* by a deterministic one (the contract). Let us state exactly what this buys: the noise on admitted negatives is no longer the model’s incompetence rate, but the **false-admit rate of the admission predicate itself** — the episode confound, defined and bounded in §4.5, audited, recoverable by retraction. That rate is not zero. And no "noise-free case" rate is restored, because §4.2 provides none: identification in the limit is qualitative, and the reference PAC bound assumes an i.i.d. the stream violates.

That is the whole trick, and it is deliberately non-statistical.

The filter has a cost, which must be faced. Discarded (non-localized) episodes concentrate on rare, hard sorts; those sorts therefore collect their clean negatives *more slowly*. The competence bias thus reappears — but as a speed cost, never as an unsound admission — and it is the optimistic half of §4.5 that keeps it from becoming self-sealing again.

Attribution is realized by two mechanisms, and their distribution is worth naming because the three grains distribute over them (formal *duality*, for its part, is reserved for the credit/blame pair of §4.6):

- (α) **Structural provenance** — the atom carries a label naming its unit (per-slot postcondition

provenance, minted when the structure is declared). Where the structure can name the unit, attribution is a lookup.

- **(β) Counterfactual ablation** — the verdict changes iff the unit changes: $\text{verdict}(P) \wedge \neg \text{verdict}(P \setminus \{u\})$, evaluated deterministically on the lattice-pure path, fallbacks disabled (a redundant fallback path would mask decisiveness). The spirit is Pearl’s intervention; the guarantee, though, is a program-analysis statement — a deterministic single-case counterfactual, not a distributional do-effect.

Slot restrictions use α ; *isa*-edge admission uses α plus verification; alias admission uses **both** — α as forward credit provenance, β as the per-unit decisiveness test. α and β have different soundness bases (a label minted at declaration; a decisiveness exercised at the episode): they are two attribution mechanisms discharging the same admission predicate. The unification claim sits at the level of the **invariant** (unique attribution $\wedge$ verification), never of the implementation.

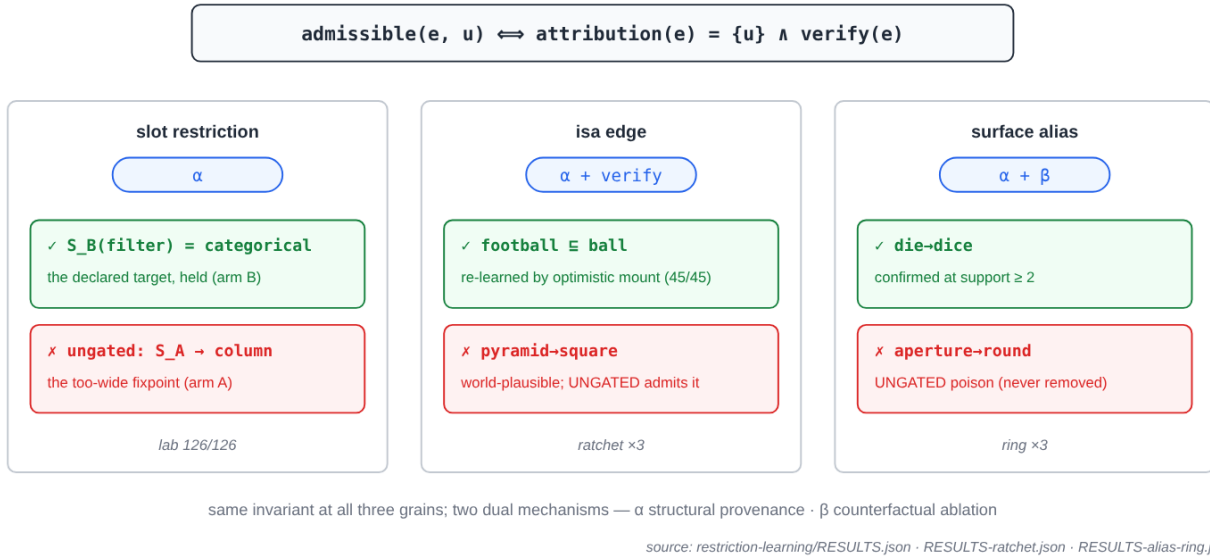


Figure F5 — the gate at its three grains: the same admission predicate, with, for each grain, a unit actually admitted and a unit actually refused in our runs — the cut held by arm B against arm A’s drift (lab), the re-learned edge against **pyramid→square** (ratchet), the confirmed alias **die→dice** against the poison **aperture→round** (ring, §6.6).

4.5 What "sound" means here: recoverability, and its two-tier envelope

The remedy is in place; it remains to say what "sound" can still mean. A per-episode admission test cannot be sound in the strongest sense, and we do not claim it. The residual hole is called the **episode confound**, and it is defined thus: a false unit can pass a single counterfactual test when the episode’s correct answer is fixed by an orthogonal factor that the false unit happens to reproduce by accident. Checking agreement is not enough to rule it out, because the proposal and its fit to the verdict come from the same correlated source — the LLM’s world-model. The envelope that bounds this hole is two-tiered, and defeasible:

- **Provisional at support 1**: admitted, but quarantined — never load-bearing for scoring a fresh episode;
- **Confirmed at support ≥ 2** over episodes with *different* orthogonal factors (de-confounded

re-exposure), via verified-reuse credit — and *confirmed* still means *defeasible*: a later localized blame retracts;

- **Blame admissibility mirrors credit admissibility**: an episode with a co-present unknown cause (an OOV hole in the very structure being scored) localizes nothing, so it can neither credit nor blame — without this rule, a *correct* learned alias oscillated admitted $\leftrightarrow$ retracted under paraphrase attrition in our runs (§6.6); that is how the rule earned its place in the library rather than in a footnote.

One limit of this envelope must be promoted to the rank it deserves: **it is the open problem of this admission**. De-confounded re-exposure neutralizes *independent* confounds — two episodes whose orthogonal factors differ do not reproduce the same accident. But §4.3’s adversary is *systematic and competence-correlated*: a bias the model reproduces on every episode (proposal and verdict-fitting come from the same world-model) can pass confirmation at support ≥ 2 . Our empirical zeros show that this systematic-confound rate is *low on these streams*; they are not a bound, and they read by the rule of three (Hanley–Lippman–Hand): zero false over N exposures bounds the rate at $< 3/N$ at 95% — never at zero. *Strong* de-confounding (a verifier independent of the proposer, from a different model family) is the first item of future work.

Soundness, as sold in this article, is therefore **recoverability**: no admission is irreversible, every admission is auditable back to the episode that produced it, and the poisoning path that requires no further interaction — the NELL signature — is closed. The self-sealing direction (§4.3) is handled by the dual policy: **optimism under uncertainty** — rejected sorts are re-mounted at doubling horizons, so a false blame costs extra mounts rather than a truth expelled forever. §5 prices both policies.

4.6 The dual of credit

Last question of the formal layer: does the same reasoning hold on the success side? Blame is only half the rule. Let us first name the case that traps credit: the **zero-side** success — the episode’s global verdict passes while one of the slots worked on zero data (its filter matched no rows). That slot has demonstrated nothing about its sort. Naive credit, which credits every slot of a composite on each global success, nevertheless generalizes $\mathbf{S}$ over it: it over-credits exactly where nothing was exercised. The dual rule — **positive credit only for roles whose atoms carry exercised provenance** — is the same localization at opposite polarity, with a deliberate asymmetry: credit is per-atom (a verified atom is direct evidence for its own role), while blame requires unanimity (a failure remains a disjunction of causes until it is localized). §5.3 prices this dual: the naive arm silently admits unverified generalizations, exactly at the zero-side successes; the localized arm pays a transient lag (one arrival per zero-side event) and converges to the evidence cut.

5. The deterministic laboratory

Before any model enters the loop, the learning rule itself goes on the bench: pure code, zero GPU, exact pre-registered expectations, permuted streams. The question is not whether the rule *works* — §4.2 says the LGG alone identifies in the limit — but what the gate *buys and costs* against its unsound alternatives, cell by cell, never on average.

Because the article compares many arms, here they all are, one line each:

Arm	What it is	Where
A / B / C / D	deterministic lab: LGG-only / + gate / + every-failure-negative / B + optimism	§5.1–§5.2
P-glob / P-loc	credit lab: success credits everything / per-atom localized credit	§5.3
SYS	the full typed path (the model extracts, the lattice decides)	§6–§7
ABLATED	SYS on an amputated lattice: must fail closed, then learn through the gate	§6.2
DIRECT	the model alone, same prose, reasoning budget 0 or 1024 (written rb)	§6–§7
RAG	DIRECT + the full declared ontology in the prompt	§7.1
UNGATED / GATED	growth without the gate (the model writes the unit) / through the gate	§6.5–§6.6
SYS-extract	DeFAB: prose → model extraction → then the typed selector	§7.3

5.1 The apparatus

The objects first: two declared lattices, with branching ≥ 3 under every target cut and one deliberate multi-parent pair (to exercise §4.1’s parallel cuts); three stream permutations; two noise regimes. The heart of the apparatus is then that each episode returns **two divergent oracle atoms**: a *permissive* surface success (a bad filter can match rows by accident — the false positive that lifts an LGG) and a deep per-slot contract, which *localizes*. This divergence is necessary: if the admission oracle and the blame contract always said the same thing, all arms would see the same stream, the wedge between arms could not exist, and the experiment would be empty. Finally the noise, two channels, both **rare-correlated** by design: N1 flips the first occurrence of each rare sort into a non-localizable failure ($\propto 1/\text{frequency}$ — §4.3’s competence profile); N2 flips blame onto the *good-rare* sort of the other slot (false blame at rate ρ). The learner reads only the pass/fail atoms — never the type table.

Four arms on bit-identical streams:

Arm	Rule
A	LGG-only (positives alone — §4.2’s baseline)
B	+ negatives admitted only if blame-localized (the gate)
C	+ <i>every</i> failure admitted as a negative (the unsound control)
D	B + optimism (doubling-horizon re-mount of rejected sorts)

5.2 The results — 126/126 exact pre-registered checks

Four dynamics were predicted; all four are measured in every cell:

1. The wedge exists and is worth its theoretical bound.

A’s over-generalization errors: **16** (every bad arrival — the LGG alone never brakes). B: **8** at $\rho=0$ — exactly one unavoidable first exposure per (facet $\times$ bad sort), the floor. The gate buys **–50% over-generalization at zero over-tightening** (good tasks refused by B: 0, identical to A).

2. A converges — to the wrong fixed point.

In every cell, $S_filter(A)$ settles on *column* (lifted there by the permissive oracle’s accidental matches); $S_filter(B)$ holds **categorical**, the declared target. The wedge is not "A does not learn"; it is "A learns *too broad*, stably" — which is worse, because it looks like convergence.

3. The unsound control self-seals, monotonically, on the rare.

C refuses 2 to 14 *good* tasks per cell, concentrated exactly on the rare sorts N1 hit — and never recovers them. B at $\rho=0$: zero. This is §4.3’s asymmetry made visible, and it is the measured answer to "why not just count every failure as a negative".

4. False blame degrades B; optimism recovers it, at a visible premium.

At $\rho > 0$, B seals good-rare (3 per cell); D ends with $\leq$ sealed (2–3) at the price of 2 extra mounts and an over-generalization of 5–6 versus 4. The insurance is not free — and at $\rho = 0$, D $\equiv$ B: the premium is only paid when there is something to insure.

One unforeseen dynamic is recorded, because it will return in the live metrics: at $\rho > 0$, B's over-generalization *goes down* (8→4) — a bad seal on a good-rare collaterally refuses the bad events that carry that sort on the other slot. A false blame can "protect" by accident while inflating over-tightening.

Moral: the two error counters are always read together; either one, alone, can be flattered by the other's failure.

5.3 The dual of credit, priced — 108/108 exact checks, 18 cells

Same discipline, opposite polarity. The streams are built with **zero-side successes** — global PASS while one slot matched nothing (the signature actually observed under live conditions: a comparison episode with one side empty because the requested entity is absent from the data). Arms: P-glob (the success credits both slots) versus P-loc (credit only through exercised per-atom provenance); the blame gate is active and identical in both arms — which is itself a result: **credit poison cannot be caught back through the blame channel** on these streams. Results: P-glob's unverified admissions equal the number of zero-side events, and its S lifts toward the too-broad cut; P-loc admits nothing undue and converges to the evidence cut, at the price of a transient lag of one arrival per zero-side event, with equal endpoints as soon as the first exercised evidence arrives. The control stream without zero-sides makes the arms bit-identical — it is the divergence, not the mechanism, that creates the wedge, exactly as in §5.1.

5.4 The failure envelope of the gate itself — deterministic control, 12/12

Finally, the gate itself is attacked deterministically: a forced-confound episode admits a false alias (§4.5's hole, made real); the envelope then retracts it, following the full cycle — provisional admission, then localized blame, retraction, unblocking, and re-admission of the displaced correct unit. The per-unit counterfactual admits the load-bearing alias and refuses the one that is empty-in-episode; a masking fallback is unmasked by lattice-pure scoring. All twelve transitions land as pre-registered. The confound hole is *real*; the claim survives because the envelope bounds it — which is precisely §4.5's recoverability semantics, exercised.

6. Existence under live conditions

The laboratory prices the rule; the live level asks whether the whole circuit — extraction, canonicalization, lattice decision, refusal, defeasance, and the two grains of learning — holds together when a real model supplies every noisy input. A single embedded model plays every role (extractor, paraphraser, proposer, and the DIRECT baseline): an open-weights 27-billion-parameter model quantized to 2 bits, run locally, reasoning budget 0 unless stated (written rb). Every episode is durably memoized: all results replay bit-for-bit. The path: the protocol (§6.1), constancy and the two kinds of learning on fresh instances (§6.2), oracle hardening (§6.3), volume over three domains (§6.4), then the two drift baselines — the edge ratchet (§6.5) and the alias ring (§6.6). The generality of these results *across extractors* is measured separately, on nine models, in §7.4.

6.1 The protocol

The task family of §3: placement riddles over declared toy lattices, in four variants that each isolate a mechanism, plus a fifth for defeasance:

- **V1** facet distractor — a salient, irrelevant color;
- **V2** depth-2 *isa* — the prose names only the subtype: unsolvable without the chain, the lattice is load-bearing by construction;
- **V3** no-match — the typed-refusal cell;
- **V4** axis product — shape alone is ambiguous, size discriminates;
- **V5** the defeasible modifiers, with their benign controls.

Three arms run them:

- **SYS** — the typed path: the model only extracts, the lattice decides;
- **ABLATED** — the sort’s edges removed: must fail closed, then learn them through the gate;
- **DIRECT** — the model answers alone, same paraphrased prose.

The prose is always paraphrased by the model — the surface is never ours. Where the oracle is the declared lattice itself, the circularity is owned and stated: those cells measure the *division of labor* (extraction $\times$ match $\times$ fail-closed $\times$ learning), not world knowledge; the external oracle lives in §7.

6.2 Constancy, defeasance, edge learning

On fresh instances and on a full domain transposition (animals/enclosures: `trout` $\sqsubseteq$ `fish` $\sqsubseteq$ `aquatic`, an aviary, a terrarium):

cell	SYS	ABLATED (strict path)	DIRECT
shapes V1–V4, fresh (24)	24/24	fail-closed 18/18 $\rightarrow$ learned and solved 18/18	20/24
shapes V5 defeasible (3)	3/3 retracted	—	1/3 (2 hallucinations)
shapes V5 benign (3)	3/3 mounted	3/3 $\rightarrow$ 3/3	3/3
animals V1–V4, transposed (24)	24/24	fail-closed 18/18 $\rightarrow$ learned and solved 18/18	16/24

(Counting convention, ABLATED column: the V3 no-match cell ablates nothing — there is no edge under a refusal — so each domain exposes 18 ablatable edges across its 24 V1–V4 tasks.)

Three readings come out of this table. Constancy: SYS scores 54/54, zero structural hallucinations, on fresh instances as on a fully transposed domain. Learning: the ablated arm recovers **39/39** missing edges across probes 1–2 (18 shapes + 18 animals + 3 benign) via the optimistic-mount + localized-credit circuit, and every pre-learning failure is closed — never a wrong hole. Defeasance: DIRECT pattern-matches through "deflated" in 2 cases out of 3, while the typed path extracts the condition 3/3, defeats, retracts — and the benign controls defeat nothing, so the discrimination is real.

The honest reading *in favor* of the baseline deserves its own sentence: DIRECT scores 40/54 on this table (42/54 on the memo-served replay — inter-process nondeterminism, §8; the shipped memo replays 42), so the model alone is almost as knowledgeable.

The typed path’s value concentrates precisely where knowing is not the question: refusal, fidelity, auditability and learnability.

6.3 Hardening the oracles — and what it revealed about both regimes

A first reading of V3 ("the model hallucinates on no-match") did not survive our own scrutiny, and the correction is more interesting than the original cell. With reasoning on, the model's V3 answers are *world-plausible*: a canonical pyramid does have a square base ("it fits base-first"); ferns do live in terrariums. Those oracles were contestable; we hardened them until no world-plausible escape remained — and the collapse of the artifact is reported as such:

hardened cell	SYS	DIRECT rb=0	DIRECT reasoning
V3h-shapes (square hole removed)	6/6 none	6/6	6/6 — the old V3's "hallucinations" were the <i>oracle's</i> artifact
V3h-animals (terrarium removed)	6/6 none	4/6 (2 hallucinations)	4/6 (2 hallucinations)
V5h-remap (deflated: round→flat slot, positive gold)	9/9 (6 retract+remap · 3 benign)	3/9	3/9

Two results carry into every later section. First, the refusal cell that survives (V3h-animals) is non-empty and incontestable: every escape removed, the direct model still fabricates a placement in 2 cases out of 6 — **in both reasoning regimes** — where the typed path refuses 6 out of 6. Second, and more generally: reasoning repairs the model's *knowledge* cells, not its *fidelity* cells. The clean statement of the contrast is not "the model hallucinates"; it is **"the model follows world-plausibility; the typed path follows the declared specification — in both regimes."** This reframing strengthens the governance reading of the claim (fidelity to a declared ontology is what a regulated deployment needs), and it is the honest reframing.

6.4 Volume and a virgin domain — 300/300 versus 245/300

Scaled to n=24 per cell over three domains — the third (plugs/sockets: `europug` $\sqsubseteq$ `round-pin` ..., defeater "bent", world-aligned by §6.3's lesson) onboarded from scratch — with exact 95% Clopper–Pearson intervals:

16 cells, n=300	SYS	DIRECT rb=0
total	300/300 [99–100%]	245/300 [77–86%]

The direct model's deficit concentrates exactly on the claim's cells: animals-V3 no-match **0/24 with 24 hallucinations**; V5-defeasible 1/3 and 0/3 on the defeater domains; plugs-V2 12/24 (the model does not know the pin geometry the declared lattice states).

The third domain deserves its own paragraph, because it tests something other than volume: it was onboarded *from scratch* — ontology, vocabulary and tasks written for the occasion — and this onboarding paid three instrument-grade findings, each folded back into the library. One: an ambiguous containment must stay OOV (a bare hypernym surface used to snap onto an arbitrary sort and *overwrite* a correct explicit category; fail-closed settled it). Two: multi-word sorts require the domain to declare its sort hints at intake. Three: scoring goes by hole identity, never by position index. We report them for a precise reason: all three failures fell exactly into the classes the

architecture predicts — surface vocabulary toward the ring, structure toward the canonicalization barrier, and never a silent wrong mount. That is evidence about the *shape* of the design, not just its numbers.

Two reservations deserve to be said. First, V5 stays at $n=3$ per cell — existence, not a rate. Second, the honest reading of perfect cells: the pooled interval aggregates 16 non-exchangeable cells (four of them at $n=3$), and a perfect cell never demonstrates a zero rate. By the rule of three (Hanley–Lippman–Hand), 0 failures out of 24 bounds the true failure rate at $\leq 12\%$ (95%, one-sided), 0 out of 6 at $\leq 39\%$, 0 out of 3 at $\leq 63\%$.

That is why the article never concludes from a single perfect cell, and reads the direct model’s deficit where it concentrates: refusal, defeasance and depth.

6.5 The ratchet — "the model writes the edge itself" versus the gate

The decisive baseline for any self-growth claim is the obvious shortcut: the model proposes the missing edge and the system takes it as is. The temptation is legitimate — §6.2 just showed that the model alone solves most riddles on its world knowledge; if it knows where the trout goes, why a gate on `trout` $\sqsubseteq$ `fish`? Two arms on identical streams, two domains $\times$ three permutations, answer:

$\times 3$ orders	UNGATED (the model writes the edge)	GATED (localization + verification)
false edges admitted	6/6 cells (pyramid→square · fern→terrestrial — the model’s world-plausible ontology, absorbed)	0 everywhere
silent downstream damage (wrong answers at 0 calls, uncorrectable)	1–2 per cell, order-robust	0 (residual: 1 in 2 cells — the vocabulary hole below)
premium		+1 to +3 calls per cell (quarantines, refusals, one retry)
amortization	≤ 1 proposal/sort ✓	identical ✓ — correct edges admit and serve the same

The ungated arm reproduces the NELL signature in miniature: plausible-but-false edges enter, then answer later episodes at zero calls, with no correction channel left — a *silent* damage, the kind no downstream check ever sees. The gate admits zero false edges on the same streams, at a bounded premium, and — the architectural point — correct edges admit and amortize identically: the gate is a ratchet, not an average.

Its single residual damage traces back to a defeater word arriving as an unforeseen synonym ("melted" $\rightarrow$ "liquefied") — the fourth independent occurrence of the same need in one evening (sorts, hole names, facet words, defeater conditions) — which is exactly the motivation for extending the gate to the vocabulary grain.

6.6 The learned alias ring — the same gate at the vocabulary grain

The full circuit reads in five steps:

1. a surface variant arrives OOV, and it is *exogenous* — the source prose carries it, since surfaces are extracted verbatim (the ring canonicalizes, never the model);
2. the OOV becomes an alias proposal, phrased context-free in the facet’s language, with no constrained grammar;
3. the **per-unit counterfactual intervention** decides, deterministically and at zero calls — u is admissible iff $\text{verdict}(P) \wedge \neg \text{verdict}(P \setminus \{u\})$, scored on the lattice-pure path, fallbacks disabled;
4. admission is provisional, then confirmed at support ≥ 2 over verified reuses;
5. a localized, OOV-free blame retracts.

Fifteen episodes, three stream orders:

×3 orders	GATED	UNGATED
admissions	exactly the 6 spec-true aliases the stream produces	the same 6 + 3 poisons
false aliases admitted / confirmed	0 / 0	3 (aperture/cavity/hole→round), never removed
task resolution (ok / typed refusal / wrong)	13 / 2 / 0	13 / 0 / 2
cost	14 proposals/arm, ≤1 per (key, token), re-proposals blocked by memo	identical

EDGE grain (ratchet, order 0, both domains):



VOCABULARY grain (ring, order 0 — 15 exposures per arm):



resolution: GATED 13/2/0 · UNGATED 13/0/2 (ok / typed refusal / wrong answer)

source: RESULTS-ratchet.json + RESULTS-alias-ring.json (shapes/order0, trace per episode)

Figure F6 — the ratchet, generated from the real traces (order 0): on top the edge grain (UNGATED absorbs **pyramid→square** and **fern→terrestrial**, silent damage; GATED zero false edges at a bounded premium); below the ring timeline (§6.6), exposure by exposure — true admissions (▲ blue), the poison absorbed by UNGATED alone (▲ red, the "waterlogged" episodes), wrong answers (×) converted into typed refusals (◇) on the GATED side, at equal resolution 13 == 13.

Three observations give this table its weight. First, two of the six admitted aliases had been planted by nobody — the paraphrase invented them (sphere→ball, circular-aperture→round) and the gate caught them anyway; the mechanism does not depend on knowing where its work will come from. Second, the poison itself arrived through a channel nobody had planted: the model spontaneously snaps generic hole words (**aperture**, **cavity**, **hole**) onto **round** — pure world-plausibility, on an incontestable oracle (no spec anywhere says an opening is round). The ungated arm absorbs all three, in every order: two wrong downstream answers plus a permanently poisoned ring — the NELL signature at the *vocabulary* grain, under live conditions. Third, at equal resolution (13 == 13), **the gate costs zero availability on this stream and converts the wrong answers into typed refusals** (GATED's two typed refusals are exactly the two tasks UNGATED answers *wrong*) — the trade a governed deployment wants to make.

The exact scope deserves its own sentence. It is the same localized-attribution gate (§4.4), applied unchanged at the surface-vocabulary grain — the invariant, never an implementation identity. The ring targets the **exogenous** surface variance of the source prose; the facts *emitted* by the model are already canonicalized by the strong extraction prompt, and a ring there would be redundant. Admission soundness remains *recoverability*: §4.5's episode confound exists, and the two-tier envelope plus the 12/12 deterministic control bound it. One last count remains: paraphrase attrition — the re-exposures where the paraphrase does not reproduce the expected variant, so the episode

can neither credit nor blame — consumed 5 of the 15 re-exposures; it is counted and reported, never silent.

7. Baseline arms and external validity

Four objections structure this section, each answered by an arm or a campaign. "*Just let the model reason*" — §6.3 and below: reasoning repairs knowledge, not fidelity. "*Just put the ontology in the prompt*" — §7.1. "*Your oracle is your own lattice*" — §7.3. "*You only have one model*" — §7.4.

7.1 The ontology in context (the RAG arm)

The complete declared ontology (taxonomy, holes, exceptions, and the lowest-derivable rule) is placed in the prompt, marked "authoritative", on the same 54 memo-served tasks:

cell		RAG-context	DIRECT	SYS
V1/V2/V4	knowledge	36/36	32/36	36/36
(×2 domains)				
V5	defeasible + benign	6/6	4/6	6/6
V3	no-match, shapes	0/6 (3 hallucinations + 3 broken formats)	6/6	6/6
V3	no-match, animals	3/6 (3 hallucinations)	0/6	6/6
total		45/54	42/54	54/54

Let us start with the honesty: context helps *on average* ($45 > 42$). It repairs exactly the knowledge cells — V1/V2 rise to 6/6, and the exception supplied in the prompt settles V5. The claim is about where it fails, and its failures concentrate on the claim's own cells, in two distinct modes, verified on the raw outputs. First mode: context *induces derivation chatter* that breaks the answer discipline — on the pyramid cell, the model recites its derivation instead of answering in the requested format (0/6, including 3 broken formats); we had already met this family of collapses elsewhere: context added at the semantic touchpoint produces narration instead of an answer. Second mode: world-plausibility *survives* a context marked "authoritative" — the fern is placed "terrestrial" against the ontology supplied in the very prompt (3/6). The lesson of both modes fits in one sentence: even equipped with its ontology, the model is not a reliable evaluator of it.

The deterministic matcher therefore justifies itself first on the *correctness* of the governance cells. The economics come next, and they are structural. Context pays one model call per episode, forever; the typed match memoizes to zero calls, with an audit trail and localized learning. Wall-clock tells the same story: the slow item in all our measurements is the LLM-per-episode arm — up to ~12 s per call under genuine reasoning on this stack — while the typed path at steady state is memo-served in near-zero time.

And the economics generalize: an ontology carried in the prompt is a recurring cost that grows with what the system knows; an ontology carried in the library is an asset — knowledge accumulates outside the context window, and per-call context stays bounded no matter how much has been learned (measured in the companion system article [Braun 2026]).

7.2 The static-rules arm

The foil of the learning claim is the same typed path with a **frozen** lattice (the ProSLM/s(CASP) configuration). On full declared lattices it is simply SYS: 54/54 — the soundness story of static pipelines, reproduced. On the ablated starts it stays at zero *by construction* — closed forever — where the gated circuit recovers 39/39 edges and the tasks behind them (§6.2). The delta between these two lines *is* contribution (c), isolated: everything else in the two arms is the same code.

7.3 External validity — the typed path on DeFAB

Both riddle oracles are ours; the external test is not. **DeFAB** (Cooper & Velasquez 2026) provides defeasible-reasoning instances whose gold labels are certified by the authors’ polynomial-time solver — a third-party, machine-verifiable oracle with zero circularity with our golds. Their headline number: rule-based solver 100%, against 65% for frontier LLMs — degrading to 23.5% under rendering variation.

We use its Level-3: defeater abduction. An instance gives a theory, an anomaly (a conclusion the rules predict but that should not hold), and six candidate exception rules, typed — one gold and five distractors (too-broad cut, wrong head, irrelevant, positive, wrong condition). The right one must be chosen. This task maps directly onto our gate: the right candidate is the one that is *load-bearing* (the ring admission’s decisiveness test), on the *right slot and right polarity*, **conservative** (it kills no preserved expectation of any covered individual — the vacuity/over-generalization tooth) and *minimal* (fewest posited facts, then coverage). The residual formal ties (≤ 2 verified candidates) go to the model — §3’s layered doctrine, at the knowledge frontier only.

arm (N=35, three domains)	total	notes
SYS (gate + in-set tie-break)	34/35	30/35 purely structural, at zero model calls
SYS-extract (prose → extraction → gate)	33/35	extraction noise costs one cell, fail-closed
DIRECT rb=0	30/35	
DIRECT reasoning	30/35	reasoning moves the errors around; it does not fix them

Attribution matters more than the margin: **all five DIRECT losses are over-general cuts** (**no_novel** ×3, **broad** ×2 — defeaters that would kill the preserved default of *another* covered individual; the per-instance typing of the direct model’s choices is replayed from the memo and shipped in the artifact with the deposit).

That is exactly the error class the gate’s conservativity tooth forbids *by construction* — the article’s mechanism, measured on someone else’s oracle. SYS’s single loss is a near-synonym pair *by benchmark design*; we report it as is — retouching the prompt to flip one cell would be gold fitting.

On Level-2 (374 dev instances), the typed selector scores **374/374 at zero calls** — their symbolic solver’s regime — while the published frontier number is 77.2%, and 19.1% under their hardest rendering modality; on our clean rendering, even the local 27B saturates (30/30 sampled), which places the frontier failure in **rendering variance, not knowledge** — precisely the exogenous surface variance our canonicalization barrier and learned ring target in our own pipeline. Let us state the reservations exactly: the benchmark’s first published slice ("tier 0"), selection mode (their published numbers are in generation mode — not comparable; our comparison is internal, same model, same protocol); their evaluation harness is not public at the time of writing, so the verifier

semantics were reimplemented from the instance fields and are flagged as a reproduction limit for the deposit.

7.4 The cross-model campaign — three layers, three sensitivities

A single model played every role so far — the article’s most obvious external-validity confound. We therefore replayed the entire live protocol (§6–§7), identically and memo-isolated per model, on **nine local models** covering four families (Qwen, Google, Mistral, Microsoft), three quantizations of one family ("extreme" 2-bit, 2-bit, 6-bit), two architectures (dense and mixture-of-experts with ~3 billion active parameters) and two size brackets (12 to 35 billion) — with a reproduction control first replaying the paper’s model bit-for-bit (the harness is faithful):

model quant)	(family, §6.4 SYS	§6.4 DIRECT	wrong mounts (shapes-V3)	DeFAB §7.3	pure	GATED edges	false
Qwen 27B Q2 (paper, control)	300/300	245/300	0 (24/24 re- fusals)	30/35		0	
Qwen 27B IQ2 (extreme 2-bit)	259/300	213/300	0	30/35		0	
Qwen 27B Q6	292/300	232/300	0	30/35		0	
gemma 31B Q2 (Google)	269/300	198/300	0 (after the fix, below)	30/35		0	
Ministral 8B Q8 (Mistral)	212/300	200/300	2	30/35		0	
phi-4 14B Q4 (Microsoft)	265/300	231/300	0 (after the fix)	30/35		0	
Coder 30B-A3B Q4 (Qwen, MoE)	252/300	168/300	0	30/35		0	
Qwen3.5 35B- A3B Q4 (Qwen, MoE)	271/300	192/300	0	30/35		0	
gemma-3 12B Q4 (Google, small)	79/300	50/300	0	30/35		0	

The verdict fits in three layers, with opposite sensitivities:

1. **The deterministic decider is invariant by construction — confirmed.** DeFAB structural-pure: 30/35 identical across all nine models; the L2 selector, 374/374 everywhere; the §5 lab never touches a model.
2. **The gate is probative multi-family — the strong result.** Zero false edges admitted across the nine models, while every family *emits* poison — and **the same** (pyramid→square, fern→terrestrial, plus their extras; the extreme-2-bit variant emits twice as much, the gate still holds at zero). A world-plausibility bias shared across families is exactly the case where a gate beats a choice of model.
3. **End-to-end fail-closed soundness holds across the four families — after a one-line fix reminded us who decides.** What that fix demonstrates deserves its own paragraph, below.

This fix is a result, not an anecdote, and here is what it demonstrates. On the first pass, gemma and phi-4 produced 24 wrong mounts on the shapes-V3 cell where Qwen refused 24/24 — same

pipeline, same rules; the memoized trace, replayable at zero GPU, localized the cause outside the models (the explicit-facet fallback also fired for *in-vocabulary* sorts, against §3 P4’s doctrine; Qwen only escaped it by an accident of style, it leaves the facet empty); a one-line guard — the doctrine as stated — brings all four families back to **zero wrong mounts**, with no regression. Three things are thereby demonstrated, and each carries beyond the episode. Soundness lives in the *deterministic path*, not in the model: fixing one line fixes four families — a model whose plausibility leaks cannot be fixed that way. Only an extractor from a *different family* could expose the gap between the stated doctrine and its implementation: that is the methodological argument for cross-model validation itself. And the entire diagnosis ran on the memoized trace, without a single GPU call: the typed-fact discipline paying off exactly where it promised to.

The last three rows of the table add the architecture axis and the size axis. Two mixture-of-experts models (~3 billion active parameters) hold all three layers, at middling coverage. The most instructive case is the dense 12-billion model: its coverage collapses — **79/300** — yet it produces *not one* wrong mount and admits *not one* false edge, while its ungated arm absorbs **thirty** (ten silent wrong downstream answers). The weaker the extractor, the more the gate pays; unable even to propose an alias, it fails into pure typed refusals. That is graceful fail-closed degradation at its sharpest.

This extension also forces two honest nuances. First: **the governance gap on refusal narrows with model generation**. The most recent model tested scores 6/6 on §6.3’s hardened refusal cells, in both regimes — but it stays modifier-blind on defeasible re-derivation (1/9 against the typed path’s 9/9) and keeps the over-general-cut losses on DeFAb (29/35 against 34/35): the claim’s load-bearing cells stay load-bearing. Second: the "zero wrong mounts" is scoped to the refusal/volume cells. The *re-derivation* cell under a defeater (V5h-remap, §6.3) keeps a residual that grows as the extractor weakens (down to 0/9 for the 12B) — counted and reported separately, never silent.

Let us state the exact scope: **end-to-end SYS soundness is conditioned on extraction** — the lattice decides correctly *given* the extracted facts, the gate and the decider are invariant, and what varies with the extractor’s capability (and quantization) is **coverage**, which degrades gracefully and fail-closed (Qwen 300 > gemma 269 > phi-4 265 > Ministral 212 > gemma-12B 79; the Qwen IQ2→Q2→Q6 ladder produces no wrong mount at any rung).

This campaign strengthens generality *across extractors* — families, quantizations, architectures, sizes; it says nothing about generality *across domains and tasks* — §8’s limit, unchanged.

8. Limits, threats, and the governance reading

The scale of the evidence. Every live result is an existence result: declared toy lattices, N=6–24 per cell (V5: n=3), one family of task schemas per domain. The §7.4 campaign extends validity *across extractors* (four model families, three quantizations of one family); it does not extend validity *across domains and tasks*, which remains the primary limit. Part of a given extractor’s "incompetence noise" may moreover be quantization damage rather than capability — the Qwen extreme-2-bit→6-bit ladder varies coverage at constant soundness, which bounds that part without eliminating it. The volume section (§6.4) gives tight intervals *within* this task family; nothing here estimates cross-family rates. The population-scale study (vocabulary growth laws over hundreds of episodes) is a separate campaign, not claimed by this article.

Oracle circularity, where it exists. The riddle cells whose oracle is the declared lattice measure the division of labor, not knowledge; this is said at every table. The cells that carry the external

claim are the hardened refusal cell (§6.3), the drift baselines of the ratchet and the ring (§6.5–6.6, where the oracle is the declared spec against which drift is defined), and DeFAB (§7.3, third-party oracle).

Admission soundness is recoverability. The episode confound (§4.5) is a real hole of single-episode counterfactual admission; the two-tier envelope bounds it and the deterministic control exercises the full poison→retraction→recovery cycle, but a unit that never re-presents de-confounded can persist in the provisional state. We count the "admitted, never re-exercised" as an at-risk compartment, rather than pretending it is empty.

Attrition and nondeterminism. Paraphrase attrition consumed re-exposures in the ring runs (5/15 — counted, reported); confirmation support is therefore conservative. Model outputs vary across processes on this GPU stack; all comparisons are within-process with durable memoization, and replays are bit-exact given the memo.

The costs not lifted. The lattice must still be declared before it can grow — the ontology-authoring cost is reduced (edges, aliases and, in the larger host system, frames are learned), not removed. The gate’s premium is bounded and measured on these streams (+1 to +3 calls per cell; zero availability cost on the ring stream); adversarial streams could price it differently.

The governance reading. §6.3’s reframed contrast — the model follows world-plausibility, the typed path follows the declared specification, in both reasoning regimes — is, we believe, the durable value proposition. A deployment that must answer "*why did the system say no?*" gets:

- a typed refusal naming the missing requirement;
- an admission audit of every learned unit, back to the episodes that earned it;
- a retraction path for every unit that stops earning it;
- and a knowledge substrate that grows with use without absorbing its extractor’s ontology — accumulating outside the context window, at bounded per-call context, so working memory never saturates, however long the deployment runs.

None of this requires the model to be honest about its own knowledge — only the gate to be strict about attribution.

9. Conclusion

Candidate elimination has been sound and fragile for forty years; large language models are knowledgeable and unfaithful today. The composite between the two — an LLM that only extracts and proposes, a typed *isa* lattice that decides, and an admission gate that admits a generalization only when its evidence is uniquely attributable and verified — turns out to be more than the sum. The lattice keeps the model’s fidelity failures out of the *answers*: 300/300 against 245/300, the refusal cells at 6/6 against 4/6 in both reasoning regimes. The gate keeps them out of the *knowledge*: zero false edges and zero false aliases where the ungated ratchet absorbs the model’s ontology and propagates silent damage — NELL’s drift, reproduced in miniature then blocked, at both grains, with identical poison across four model families. The mechanism is priced in a deterministic laboratory (–50% over-generalization at zero over-tightening; the unsound control self-seals; localization is what makes the difference, on the blame side as on the credit side), and it survives a third-party oracle, where every loss of the direct model is precisely the over-general cut the gate’s conservativity tooth forbids.

One gate, three grains: the restriction, the edge, the alias. What was missing between a knowledge base that cannot learn and a model that cannot be trusted was never more intelligence — it was

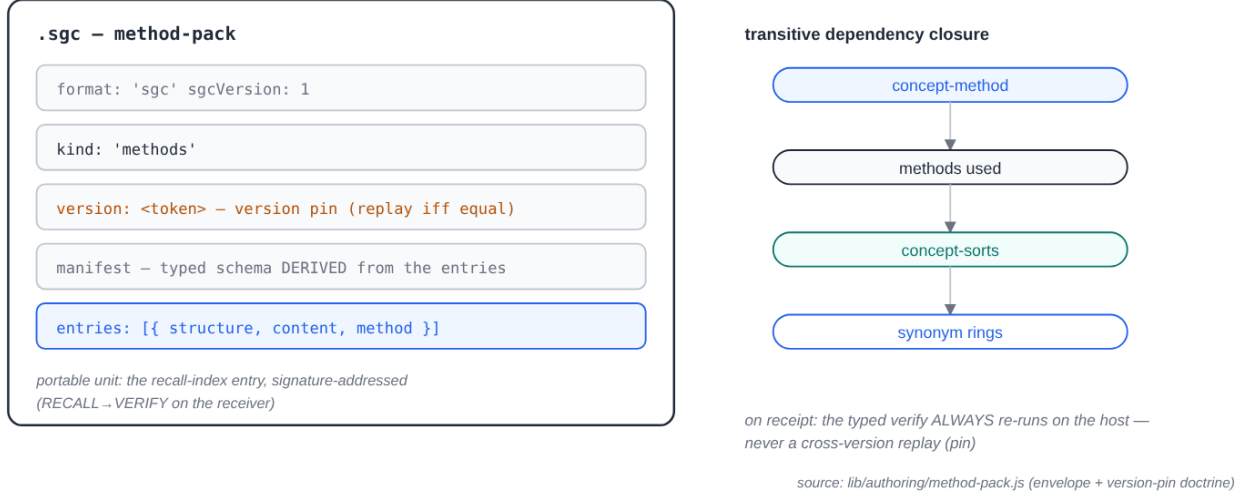


Figure F8 (host context, [Braun 2026]) — the substrate’s export and audit unit: the `.sgc` method-pack ships a concept-method with the transitive closure of its dependencies — used methods, concept-sorts, rings — its contract and its provenance; the version pin forbids cross-version replay, and typed verification always re-runs at the receiving host. No claim of the present article rests on this figure.

an admission rule.

Appendix A — The figures (generated and integrated)

All eight figures are **generated from the recorded artifacts** of the real runs (the result JSONs, the durable memo, the lattice declaration in the probe source, the final rings, the computed LGGs) by the script `figures/generate-figures.js` (zero dependencies, shipped with the deposit) — never redrawn from memory. Every SVG embeds its provenance as a comment. Regenerate: `node figures/generate-figures.js`. Reading note: the prose examples in the body are English renderings; the figure tokens are the actual surfaces the model emitted in the runs.

- **F1** (§3 P1) — a paraphrased sentence, annotated: prose spans → typed facts; object and slot indicators. Source: an intake trace from the riddle RESULTS.
- **F2** (§3 P3, §4.1) — the shapes lattice with a restriction drawn as a cut; defeasible edges marked {source, confidence}. Source: the declared lattice + a learned-edge record.
- **F3** (§3 host) — anatomy of a concept-method: subpath forest, role-typed slots, parallel qualification segments. Source: a method crystallized from the parametric-reuse artifacts.
- **F4** (§3 P4/P5) — the mount: successful (params→slots, zero firing) versus starved (→ typed hint). Source: mount traces, both outcomes.
- **F5** (§4.4) — THE gate, three columns (slot / edge / alias): same invariant (unique attribution ∧ verification), attribution mechanisms α/β marked per grain. Schematic, but the three example rows are real admitted units.
- **F6** (§6.5–6.6) — the ratchet timeline: UNGATED absorbing false units and answering wrongly at zero calls, versus GATED refusing, quarantining, retracting. Source: RESULTS-ratchet + RESULTS-alias-ring, per episode.
- **F7** (§3 P2, canon) — an affinity/digram fold: [filter→aggregate] folded by MDL admission. Source: the fold’s admission record (ΔL , support).

- **F8** (§8) — a method-pack export: a concept-method with the transitive closure of its dependencies (methods, sorts, rings) + contract + provenance. Source: an `.sgc` pack.

Appendix B — Reproducibility

The main runs use an embedded open-weights model (27 billion parameters, 2-bit quantization), reasoning budget 0 unless stated (reasoning arm: budget 1024, answer budget sized above the thinking budget — an early arm that starved the answer is reported and discarded in the logs). The §7.4 campaign replays the protocol on the other models with the same harness, each model isolated in its own result artifacts, the durable memo shared and keyed per model; the inference backend is the official upstream implementation, KV cache at full precision — audited at the live-process level to rule out any modified build. Every model call is durably memoized (content-addressed); every table replays bit-for-bit from the shipped memo. The deterministic laboratory and the control experiments are pure code (zero GPU). The DeFAB data is public (MIT); the authors’ evaluation harness was not available at the time of writing — our verifier semantics is reimplemented from the instance fields and flagged as such (§7.3). [Note: artifact-bundle structure + DOI to be finalized at deposit.]

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